

$$\begin{aligned}
& |\varphi_\varepsilon * f(x) - f(x)| \\
&= \left| \int_{\mathbb{R}^n} [f(x - \varepsilon y) - f(y)] \varphi(y) dy \right| \\
&\leq \int_{\mathbb{R}^n} |f(x - \varepsilon y) - f(x)| |\varphi(y)|^{\frac{1}{p}} |\varphi(y)|^{\frac{1}{q}} dy \\
&\leq \left[\int_{\mathbb{R}^n} |f(x - \varepsilon y) - f(x)|^p |\varphi(y)| dy \right]^{\frac{1}{p}}.
\end{aligned}$$

对上式两端作 p 次乘方再对变量 x 作积分, 运用 Fubini 定理得

$$\begin{aligned}
& \int_{\mathbb{R}^n} |\varphi_\varepsilon * f(x) - f(x)|^p dx \\
&\leq \int_{\mathbb{R}^n} |\varphi(y)| \left\{ \int_{\mathbb{R}^n} |f(x - \varepsilon y) - f(x)|^p dx \right\} dy.
\end{aligned}$$

令 $\varepsilon \rightarrow 0$, 因为

$$\int_{\mathbb{R}^n} |f(x - \varepsilon y) - f(x)|^p dx \leq 2^p \|f\|_p^p,$$

由控制收敛定理, 上面不等式右端可在积分号下取极限, 再运用积分平均连续性 (引理 4.3.2) 得

$$\begin{aligned}
& \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n} |\varphi_\varepsilon * f(x) - f(x)|^p dx \\
&= \int_{\mathbb{R}^n} |\varphi(y)| \left\{ \lim_{\varepsilon \rightarrow 0} \int_{\mathbb{R}^n} |f(x - \varepsilon y) - f(x)|^p dx \right\} dy = 0.
\end{aligned}$$

在 $L^1(\mathbb{R}^n)$ 中虽然卷积运算无单位元, 但是 $\varphi_\varepsilon * f \xrightarrow{L^1} f$, 所以 φ_ε 称为卷积运算的渐近单位元.

例 1 考虑函数 $\varphi(x) = \chi_{[-1,1]}$, 则

$$\varphi_\varepsilon(x) = \frac{1}{\varepsilon} \chi_{[-\varepsilon, \varepsilon]}(x).$$

易见对任给 $f \in L^p(\mathbb{R})$, 有

$$\varphi_\varepsilon * f(x) = \frac{1}{\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(y) dy \stackrel{\text{def}}{=} f_\varepsilon(x).$$

于是由定理 4.3.1, $\|f_\varepsilon\|_p \leq \|f\|_p$; 且由定理 4.3.6, $f_\varepsilon \xrightarrow{L^p} f$.